Topics 7.1 – 7.5: MCQ Practice

$$2\ Cl_2(g) + O_2(g) \rightleftarrows 2\ Cl_2O(g)$$

1. Equimolar amounts of $Cl_2(g)$ and $O_2(g)$ are injected into a previously evacuated rigid reaction vessel, where they react according to the equation above. The partial pressures of the gases in the reaction vessel are monitored at constant temperature and recorded in the table below.

Time	P_{Cl_2} (atm)	P_{O_2} (atm)	P_{Cl_2O} (atm)
t_0	0.50	0.50	0.0
t_1	0.42	0.46	0.08
t_2	0.35	0.43	0.15
t_3	0.29	0.40	0.21
t_4	0.25	0.38	0.25
t_5	0.25	0.38	0.25

Which of the following correctly compares the rate of the forward reaction with the rate of the reverse reaction between times t_2 and t_3, and provides the correct justification?

	Comparison of Reaction Rates between times t_2 and t_3	Justification
(A)	The rate of the forward reaction is equal to the rate of the reverse reaction.	The partial pressure of the product decreases.
(B)	The rate of the forward reaction is equal to the rate of the reverse reaction.	The partial pressure of the product increases.
(C)	The rate of the forward reaction is greater than the rate of the reverse reaction.	The partial pressure of the product decreases.
(D)	The rate of the forward reaction is greater than the rate of the reverse reaction.	The partial pressure of the product increases.

$$2\ XY(g) \rightleftarrows X_2(g) + Y_2(g) \qquad K_p = 9.00$$

2. A pure sample of $XY(g)$ is pumped into a previously evacuated, rigid container and allowed to attain the equilibrium shown above at a constant temperature. If the partial pressure of $XY(g)$ is equal to 0.500 atm at equilibrium, what was the initial pressure of $XY(g)$ in the container?

(A) 1.00 atm

(B) 1.50 atm

(C) 3.00 atm

(D) 3.50 atm

$$PCl_5(g) \rightleftarrows PCl_3(g) + Cl_2(g)$$

3. $PCl_5(g)$ decomposes into $PCl_3(g)$ and $Cl_2(g)$ according to the equation above. A pure sample of $PCl_5(g)$ is placed in a rigid, evacuated container. The initial pressure of the $PCl_5(g)$ is 1.00 atm. The temperature is held constant until the system reaches a state of equilibrium. Information about the initial and the equilibrium conditions of the system is listed in the table below.

	Initial Conditions	Equilibrium Conditions
Gases Present in the Reaction Vessel	$PCl_5(g)$ only	$PCl_5(g)$, $PCl_3(g)$, and $Cl_2(g)$
Total Pressure in the Reaction Vessel	1.00 atm	1.40 atm

As the reaction progresses toward equilibrium, the rate of the forward reaction

(A) increases until it becomes the same as the reverse reaction rate at equilibrium

(B) stays constant before and after equilibrium is reached

(C) decreases to become a constant nonzero rate at equilibrium

(D) decreases to become zero at equilibrium

$$2\ AX_2(g) \rightleftarrows A_2X_4(g)$$

4. The synthesis of $A_2X_4(g)$ is represented by the equation above. A 0.700 mol sample of $AX_2(g)$ is placed in a rigid 1.00 L reaction vessel and allowed to reach equilibrium at a certain temperature. What is the value of K_c at this temperature if 0.200 mol of $A_2X_4(g)$ is present at equilibrium?

(A) 0.450

(B) 0.667

(C) 0.800

(D) 2.22

$$2\ F_2(g) + O_2(g) \rightleftarrows 2\ OF_2(g)$$

5. Equimolar amounts of $F_2(g)$ and $O_2(g)$ are placed in a previously evacuated rigid reaction vessel, where they react according to the equation above. The partial pressures of the gases in the reaction vessel are monitored at constant temperature and recorded in the table below.

Time	P_{F_2} (atm)	P_{O_2} (atm)	P_{OF_2} (atm)
t_0	1.0	1.0	0.0
t_1	0.78	0.89	0.22
t_2	0.60	0.80	0.40
t_3	0.46	0.73	0.54
t_4	0.36	0.68	0.64
t_5	0.30	0.65	0.70
t_6	0.30	0.65	0.70

Which of the following provides the correct information about the system at time t_4 ?

	Approximate Value of Q_p at time t_4	Comparison of Q_p and K_p at time t_4
(A)	2.6	Q_p is less than K_p
(B)	4.6	Q_p is less than K_p
(C)	2.6	Q_p is greater than K_p
(D)	4.6	Q_p is greater than K_p

$$COCl_2(g) \rightleftarrows CO(g) + Cl_2(g)$$

6. $COCl_2(g)$ decomposes according to the equation above. When pure $COCl_2(g)$ is injected into a rigid, previously evacuated flask at 690 K, the pressure in the flask is initially 1.0 atm. After the reaction reaches equilibrium at 690 K, the total pressure in the flask is 1.2 atm. What is the value of K_p for the reaction at 690 K?

(A) 0.040

(B) 0.050

(C) 0.80

(D) 1.0

$$2\ AX_3(g) \rightleftarrows 2\ AX_2(g) + X_2(g) \qquad K_p = 4.0 \text{ at } 500\ K$$

7. A chemist adds a pure sample of $AX_3(g)$ to a previously evacuated rigid reaction vessel. A decomposition reaction occurs at 500 K according to the equation above. The partial pressures of $AX_3(g)$, $AX_2(g)$, and $X_2(g)$ in the vessel are monitored over time, as shown in the table below.

Time	P_{AX_3} (atm)	P_{AX_2} (atm)	P_{X_2} (atm)
t_0	6.00	0.0	0.0
t_1	2.00	4.00	1.00
t_2	2.00	5.38	1.00
t_3	2.32	5.06	0.840

At time t_1, the system reaches equilibrium. At time t_2, additional $AX_2(g)$ is injected into the reaction vessel, and the temperature of the reaction mixture is maintained at 500 K.

Which of the following statements concerning the system at time t_3 is correct?

(A) Q_p is less than K_p.

(B) Q_p is equal to K_p.

(C) Q_p is greater than K_p.

(D) The values of Q_p and K_p cannot be compared without knowing the volume of the reaction vessel.

$$H_3PO_4 \rightleftarrows H^+ + H_2PO_4^- \quad K_{a1} = 7.2 \times 10^{-3}$$
$$H_2PO_4^- \rightleftarrows H^+ + HPO_4^{2-} \quad K_{a2} = 6.3 \times 10^{-8}$$
$$HPO_4^{2-} \rightleftarrows H^+ + PO_4^{3-} \quad K_{a3} = 4.5 \times 10^{-13}$$

8. A solution is prepared by mixing 50 mL of 1 *M* NaH_2PO_4 with 50 mL of 1 *M* Na_2HPO_4. On the basis of the information above, which of the following species is present in the solution at the lowest concentration?

(A) Na^+

(B) HPO_4^{2-}

(C) $H_2PO_4^-$

(D) PO_4^{3-}

$$CO(g) + H_2O(g) \rightleftarrows CO_2(g) + H_2(g) \qquad K_c = 1.5 \times 10^3$$

9. A 2.0 mol sample of $CO(g)$ and a 2.0 mol sample of $H_2O(g)$ are introduced into a previously evacuated 100. L rigid container, and the temperature is held constant as the reaction represented above reaches equilibrium. Which of the following is true at equilibrium?

   (A) $[H_2O] > [CO]$ and $[CO_2] > [H_2]$

   (B) $[H_2O] > [H_2]$

   (C) $[CO_2] > [CO]$

   (D) $[CO] = [H_2O] = [CO_2] = [H_2]$

P_{NO} (atm)	P_{O_2} (atm)
6.00	5.00

10. Samples of $NO(g)$ and $O_2(g)$ are introduced into a previously evacuated rigid container at 300 K. The partial pressures of the gases in the container before any reaction occurs are shown in the table above. The temperature is held constant at 300 K as the reaction represented by the equation shown below reaches equilibrium.

$$2\ NO(g) + O_2(g) \rightleftarrows 2\ NO_2(g) \qquad K_p = 2.4 \times 10^{12} \text{ at } 300\ K$$

Considering the value of K_p, the total pressure in the reaction vessel at equilibrium is closest to

(A) 6.00 atm

(B) 8.00 atm

(C) 9.00 atm

(D) 11.0 atm